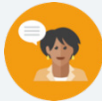


2.1 Introduction to Euclidean Geometry

Lesson Introduction

 Benchmarks	MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.		 Learning Targets	- I can distinguish between definition, postulate, and theorem. - I understand the role of proof. - I can relate properties of equality to the properties of congruence.
 Benchmark Coherence	Building On N/A		Working Toward N/A	
 Essential Questions	- What is the difference between a definition, a postulate, and a theorem? - What is the role of a proof in Geometry? - How are the properties of equality related to the properties of congruence?			
 Lesson Narrative	In this lesson, students will begin by exploring the concept of Euclidean geometry. They will be introduced to the new vocabulary “postulate” and “theorem.” Using the vertical angles theorem, students will observe how a pictorial representation, or proof, can be used to build toward more formal proofs. The lesson then moves students into formal connections of the reflexive, symmetric, and transitive properties as properties of equality and congruence. Distinction is made between the transitive property of equality and the substitution property of equality and also between equality and congruence.			
 Component Summary	2.1.1 Warm-Up (~5 min.)	Math with number cubes (<i>SE p. 83</i>)	2.1.3 Guided Instruction (15 min.)	Connections between properties of equality and congruence (<i>SE p. 86</i>)
	2.1.2 Exploration (25 min.)	Exploration into Euclidean geometry and proofs (<i>SE p. 84</i>)	2.1.4 Wrap-Up (~5 min.)	Completing a table with terms known, things wondered, and things learned (<i>SE p. 88</i>)
 Resources	Glossary		Materials	Additional Preparation
	<u>Key Terms:</u> - Postulate - Theorem - Vertical Angles Theorem - Definition of congruence <u>Additional Terms:</u> - Reflexive - Symmetric - Transitive - Equality - Congruence		(Optional) Patty paper (Optional) Protractor (Optional) Compass (Optional) Ruler	- This lesson serves as the foundation of the unit and introduces essential vocabulary. Review the vocabulary and historical figures to have a better understanding of what is going to be presented to students. - If one has not already been started, consider having the students create a proof resource booklet, notebook, or type of study guide that will continue to be added to throughout the course. This proof reference would include definitions, postulates, and theorems.

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may mistake the concept of a postulate and a theorem. - Students may mix up an equality statement and a congruence statement. - Students may not recall the correct property for a given statement.	- This lesson is heavy with introductory information and rich in vocabulary. Consider a way to capture the terms introduced in the lesson such as a guided notes sheet, foldable, or Frayer model. - Consider staggering 2.1.2 Exploration by having students stop when they get to a new vocabulary term (blue box) so it can be discussed as a whole group. - Consider creating manipulatives using card stock for 2.1.2 Exploration.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 2.1.1 Warm-Up provides students with an activity that shows a picture of 16 number cubes. Students will be asked about what they notice. Consider also asking students what they wonder to fully leverage a Notice and Wonder protocol.	MA.K12.MTR.4.1: Discussion and Reflection This lesson is an introduction to Geometry and focused on the use of MA.K12.MTR.4.1. Through the 2.1.2 Exploration, students are engaged in discussions about mathematical ideas and vocabulary. Students justify results and construct possible arguments. Teachers should establish a culture in which students ask questions and understand mistakes as an opportunity to learn.
 Differentiate Instruction	Consider pairing students who struggle with a stronger partner who can maintain collaboration so one student is not doing the primary heavy lift of the Exploration in this lesson.	
Lesson Notes:		

2.1.1 Warm-Up: Number Cubes Math

Component Overview: Students will be presented with 16 number cubes and will respond to three questions about them. For questions 2 and 3, check that students are explaining their work, as this will highlight their growth toward stronger number sense.



2.1.1 Warm-Up: Number Cubes Math

A picture of 16 number cubes is shown.



1. What do you notice?

Answers will vary. I notice that half of the dice are blue with a number 4 on them, and the other half are white with a 3 on them.

2. Find the sum of the number cubes. Show your work or explain your thinking.

Answers will vary. I counted the number of cubes with 3 dots and found that there are 8 of them. Then, I counted the number of cubes with 4 dots and found that there are 8 of them as well. I multiplied the number of dots with the number of cubes and added the products. $8 \cdot 3 = 24$ also $8 \cdot 4 = 32$ then to add $24 + 32 = 56$ so 56 is the sum of the number cubes.

3. If all of the number cubes with 3 dots were changed to number cubes with 6 dots, how would you find the new sum? Show your work or explain your thinking.

Answers will vary. If the dots on all the number cubes with 3 dots were doubled, then we can double the sum we found previously as 24 to 48. Now I add the 48 to the same 32 that we got from all the cubes with a 4 to get $48 + 32 = 80$. So, the new sum would be 80.



If time permits, consider having students share any reasonable things that they notice about the image. Don't limit the positive responses to what students notice to things that are mathematically appropriate. For example, if a student says there are blue and white dice, that is a valid observation.

Consider capturing all the things students notice on chart paper or a whiteboard and provide multiple opportunities for student voice.

2.1.2 Exploration: What Is Euclidean Geometry?

Component Overview: Students will explore key introductory vocabulary, postulates, and theorems of Euclidean geometry. Allow students to work in pairs to complete the Exploration.



2.1.2 Exploration: What is Euclidean Geometry?

Euclidean geometry is the study of planes and solid figures on the basis of axioms and theorems compiled by the Greek mathematician Euclid. Rather than using algorithms to solve problems like you may have done in other math courses, Euclidean geometry emphasizes using facts, ideas, and assumptions.



Euclid is known as the father of Geometry. He wrote 13 books using an axiomatic system to construct bodies of knowledge.

Each of Euclid's 13 books have proofs. Each proof shows how a hypothesis is true. The proof uses definitions, postulates, and proven theorems to explain why the hypothesis is true.

A **postulate** is an idea that is assumed to be true or something that is self-evident.

The first postulate in Euclid's Book 1 is shown.

A straight line segment can be drawn joining any two points.

1. Use a straightedge to show the postulate using the points shown.



For the introductory paragraph, consider highlighting keywords that may be unfamiliar to students prior to releasing the students to read. This may provide students a better understanding of the information. For example, the words "axiom," "theorem," and "proven notions" may be unfamiliar.



This exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. Allow students the opportunity to revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student misconceptions that should be addressed. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.



Students will likely not be familiar with who Euclid was or his impact on geometry. Consider having students further explore the history of Geometry, Euclid, or Euclid's 13 books.

2.1.2 Exploration: What Is Euclidean Geometry? (continued)

2. Discuss with your partner why a postulate is something that is self-evident. Summarize your discussion.

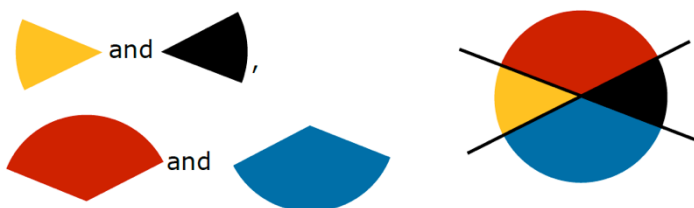
Answers will vary.

A **theorem** is an idea that has to be shown to be true through a chain of reasoning, often called a proof.

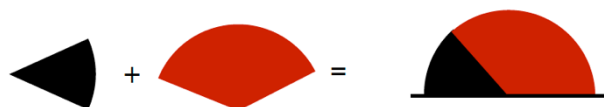
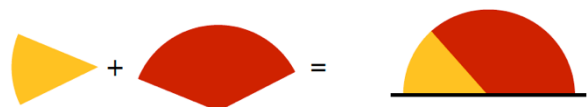


Vertical Angles Theorem: If two lines intersect, then vertical angles are congruent.

The vertical angles are



A pictorial proof of the vertical angles theorem is shown.



In the same manner,



Students may need assistance with organizing the new vocabulary of this lesson. Consider using an anchor chart, foldable, or Frayer Model for each of these terms. When reviewing the new vocabulary, it is important for students to understand the similarities between them, but more importantly, what makes them different. For example, they should see how a theorem is not the same as a postulate and why.



Students are first introduced to vertical angles in benchmark MA.8.GR.1.4. Students may struggle with the pictorial proof. Consider having patty paper, protractors, and other tools available for students to explore the proof. If students are creating a proof resource booklet, notebook, or type of study guide, add the vertical angles theorem.

2.1.2 Exploration: What Is Euclidean Geometry? (continued)

3. Discuss with your partner the pictorial proof of the vertical angles theorem below. Complete the description for each picture shown in the pictorial proof.

Picture	Description
	The yellow angle and the red angle are being added.
	<i>The black angle and the red angle are being added.</i>
	<i>This image shows the yellow and red angles being added equal 180°.</i>
	<i>The black angle added to the red angle equal 180°.</i>
	<i>The yellow angle and the black angle are equal in size.</i>
	<i>The red angle and the blue angle are equal in size.</i>

Proofs play an important role in the course of geometry. Proofs can take on many forms. They are the set of logical steps that explain why something is true.

When programmers write a new game, they use a set of logical steps. When you tie your shoes, you use a set of logical steps.



If available, consider using physical manipulatives or representations here. The images provide a visual representation of the proof, but kinesthetic learners may still struggle with the activity. By allowing students to manipulate the shapes to match the pictures, it can help provide a hands-on opportunity during the Exploration.



Questions to consider:

- What type of angle is formed by the angles in Image 3? (Straight angle.)
- How do you know that the third image shows two angles being added together add to 180°?
- How do you know that the yellow angle and the red angle add to be 180°?
- What is the relationship between the addition of the yellow and red angles and the addition of the black and red angles?
- How can you conclude that the yellow and black angles are equal in size?
- How do you know the red angle and the blue angles are equal in size?

2.1.2 Exploration: What Is Euclidean Geometry? (continued)

4. Work with your partner to place the steps in tying a pair of shoes in order.

Step	Description	Picture
2	Make a loop in one lace.	
1	Cross the laces and pull them tight.	
4	Pull both loops tight.	
3	Wrap the other lace around the loop and push some lace through the hole that is made at the bottom to make a second loop.	

5. Discuss with your partner why someone who learned to tie their shoes using a different method may have difficulty putting these steps in a logical order. Summarize your discussion.

Answers will vary. If you learned to tie your shoes in a different way, you may use steps that are not in the table.



Consider spending a few moments discussing the benefits of proofs. Students will likely struggle as they begin working with proofs, and you may hear them articulate a wondering of why they need to learn them. Beyond mathematical applications, consider sharing additional benefits such as:

- *a means of communicating your reasoning with others.*
- *a justification for the way things work.*
- *a basis for developing other applications of logical reasoning.*



Consider pausing before moving forward to the next activity. Summarize the pictorial proof for the vertical angles theorem and the logical steps for tying a shoe. Allow the Exploration to be free of the pressure of being correct but more of an Exploration of thinking.

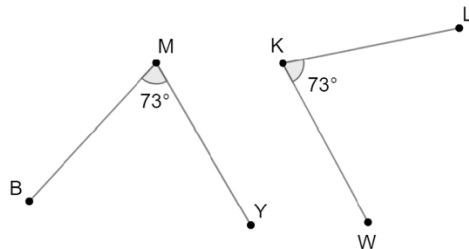
2.1.3 Guided Instruction: Properties of Equality and Congruence

Component Overview: Students will receive formal instructions on the properties of equality and congruence. Consider using gradual release during this activity.



2.1.3 Guided Instruction: Properties of Equality and Congruence

In geometry, there are different notations for congruence and equality. For example, $\angle BMY$ and $\angle LKW$ have the same measure, so the angle measurements are equal. This is written as $m\angle BMY = m\angle LKW$. It can also be said that the angles are congruent, which is written as $\angle BMY \cong \angle LKW$.



Equality and congruence may seem to be the same, but equality is used when relationships can be expressed using numbers. Congruence is used when the relationship is not noted using numbers.

- For each property, complete the example(s).

- Reflexive property

Equality	Congruence
$m\angle WHD = m\angle WHD$	$\angle WHD \cong \angle WHD$

- Symmetric property

Equality	Congruence
If $m\angle CKR = 59^\circ$ then $59^\circ = m\angle CKR$.	If $\angle CKR \cong \angle HDE$, then $\angle HDE \cong \angle CKR$.

- Transitive property

Equality	Congruence
If $m\angle CRK = m\angle HWD$ and $m\angle HWD = 90^\circ$, then $m\angle CRK = 90^\circ$.	If $\angle CRK \cong \angle HWD$ and $\angle HWD \cong \angle MPX$, then $\angle CRK \cong \angle MPX$.



This component contains the importance of proper notation for equals and congruence. Pause to ask students to consider what is the same and/or different about the two statements $m\angle BMY = m\angle LKW$ and $\angle BMY \cong \angle LKW$.

While students may better understand the substitution property and its use as a correct justification, the transitive property will be used in this course where it is appropriate. Consider addressing the level of acceptance of either property in your class when appropriate.

2.1.3 Guided Instruction: Properties of Equality and Congruence (continued)

d. Substitution property

Equality
If $m\angle CRK + m\angle HWD = 156^\circ$ and $m\angle CRK = 84^\circ$, then $84^\circ + m\angle HWD = 156^\circ$.

2. Write the conclusion for each statement. Then, name the property.

Statement	Property
If $m\angle FDT + m\angle BYH = 245^\circ$ and $245^\circ = m\angle MPW$, then $m\angle FDT + m\angle BYH = m\angle MPW$.	Transitive Property of Equality
If $m\angle FDT + m\angle BYH = m\angle RLC + m\angle KNW$ and $m\angle RLC = m\angle BYH$, then $m\angle FDT + m\angle BYH = m\angle BYH + m\angle KNW$.	Substitution Property of Equality

3. Discuss with your partner the conditional statement, "If $m\angle CRK = m\angle HWD$, then $\angle CRK \cong \angle HWD$." Summarize your discussion.

When the measures of two angles are equal, then those two angles must also be congruent.

In geometry, there is a formal difference between equality and congruence. Equality refers to measurements, which is why $m\angle CRK = m\angle HWD$ includes the m for measurement. On the other hand, congruence means the shape is the same, which can be shown using isometry. In other words, the figure coincides exactly when superimposed.



If two figures have equal measurements, then by the **definition of congruence** the figures are congruent. If congruence is shown through a series of transformations, then the **definition of congruence in terms of rigid motions** was applied.



As a possible scaffold, consider providing students with a word bank with options of properties for students to choose from to complete the table. If you do this, make sure there are reasonable answer choices and quality distractors that would require the student to know which one represents the given statement.



Students may need to be referred back to Unit 1 to recall isometry and rigid motions.

2.1.3 Guided Instruction: Properties of Equality and Congruence (continued)

The definition of congruence can be used to change from congruence to equality.

Statement	Reason
1. $\angle CRK \cong \angle HWD$	1. Given
2. $m\angle CRK = m\angle HWD$	2. Definition of congruence

Or, the definition of congruence can be used to change from equality to congruence.

Statement	Reason
1. $m\angle CRK = m\angle HWD$	1. Given
2. $\angle CRK \cong \angle HWD$	2. Definition of congruence



Gottfried Wilhelm Leibniz (1646-1716) introduced a symbol using one horizontal bar under the tilde for congruence in an unpublished manuscript of 1679. In 1824, Carl Brandan Mollweide (1774-1825) used the modern congruent symbol with two horizontal bars under a tilde in the book, *Euklid's Elemente*.



The definition of congruence can be used to state that “two figures are congruent if their measures are equal” and to state that “two figures are equal if they are congruent.” Do you think all definitions can be used “in reverse” like this? (This question sets up future learning that definitions are biconditionals.)



The historical information is here as an added historical context to the lesson. Students should not be expected to memorize names or dates related to the development of the modern-day congruency symbol.

2.1.4 Wrap-Up: Cool Down

Component Overview: Students will complete a table related to what they learned in the Exploration of this lesson.



2.1.4 Wrap-Up: Cool Down

1. Complete the table.
Answers will vary.

The terms and information used in the Exploration and Guided Instruction that I knew were...	Something I wondered during the Exploration or Guided Instruction was...	Something I learned during the Exploration or Guided Instruction was...



While this activity is open-ended, it can provide you with information on what students recall from this lesson. It can let you know what concepts and vocabulary terms stuck with students.